

**AI-Driven Crop Disease Prediction and Precision Farming  
Assistant**

**AGILE SPRINT PLANNING & USER STORY  
WORKSHOP**

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# 1. Project Vision and Stakeholders

## 1.1 Project Vision

*Vision Statement: To empower farmers, agronomists, and agricultural enterprises with an AI-driven precision farming platform that predicts, detects, and classifies crop diseases and soil stress in real-time—transforming crop protection from reactive chemical usage into an intelligent, proactive, data-driven management system that maximizes yield, minimizes environmental impact, and enhances global food security.*

## 1.2 Project Objectives

- Automatically detect plant leaf and crop diseases using deep learning computer vision models.
- Classify disease types and severity levels from mobile photo uploads, drone feeds, and multispectral field imagery.
- Provide actionable precision farming treatment guidance (dosage of biopesticides, fertilizers, and irrigation adjustments) based on localized crop conditions.
- Deliver real-time automated alerts to field operators upon identifying localized disease hotspots or pest infestations.
- Support predictive risk modeling by correlating micro-climate weather data, soil moisture levels, and historical disease outbreaks.
- Enable offline field operations on edge mobile devices for remote farming regions with low connectivity.
- Integrate seamlessly with farm management systems (FMIS) and smart IoT irrigation/sprayer hardware.

## 1.3 Key Stakeholders

| Stakeholder                              | Role & Interest in the Sprint                                                                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Product Owner (Chief Agronomist)         | Owns and prioritizes the product backlog; ensures agronomic accuracy of recommendations and accepts/rejects completed increments. |
| Scrum Master                             | Facilitates sprint ceremonies, removes impediments, coaches the team on Agile practices, and protects team capacity.              |
| Development Team (AI/ML, Developers, QA) | Designs, trains, tests, and delivers working software increments each sprint; estimates story points collaboratively.             |
| Agricultural Enterprise / Board          | Sponsors the project, sets strategic productivity goals, and evaluates sprint outcomes against ROI and farm yield metrics.        |
| Farm Manager / Field Supervisor          | Provides operational workflow requirements, validates field usability, and participates in sprint reviews.                        |

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Farmers & Field Workers          | Act as primary end-users; validate ease-of-use for mobile scanning and actionable UI notifications.               |
| Extension Officers / Agronomists | Provide domain expertise for ground-truth image labeling, disease symptom validation, and treatment verification. |
| IT & Cloud Infrastructure Team   | Manages edge AI deployment, cloud pipeline scalability, spatial database setup, and secure API gateways.          |

## 2. Epics and User Stories

### 2.1 Epic Overview

| Epic ID | Epic Name                                   | Description                                                                                                 |
|---------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| EPIC-1  | Image Acquisition & Edge Preprocessing      | Capture and prepare leaf/crop images via mobile app, drone, or satellite for AI inference.                  |
| EPIC-2  | AI-Powered Disease Detection & Localization | Apply computer vision models to identify signs of crop disease, pests, and nutrient deficiencies.           |
| EPIC-3  | Severity Scoring & Treatment Engine         | Measure severity percentage and generate targeted treatment plans (pesticides, biopesticides, fertilizers). |
| EPIC-4  | Real-Time Monitoring & Alerts               | Track field health on geospatial dashboards and issue threshold notifications for high disease risks.       |
| EPIC-5  | Analytics, Yield & Historical Reporting     | Generate farm performance reports and historical disease trend analysis for predictive planning.            |
| EPIC-6  | System Admin, Offline Mode & Integration    | Manage user roles, secure data, enable offline mobile sync, and integrate with IoT sprayers/FMIS.           |

### 2.2 User Stories by Epic

#### EPIC-1 — Image Acquisition & Edge Preprocessing

- **US-01:** As a farmer, I want the mobile application to automatically capture focused leaf images when framing a crop, so that I can inspect plant health without adjusting manual focus settings.
- **US-02:** As an IT/Mobile Engineer, I want the app to preprocess images at the edge (cropping, glare reduction, normalization), so that bandwidth usage is optimized and clean images are sent for analysis.

- **US-03:** As a system administrator, I want to configure resolution and lighting thresholds for different device camera types, ensuring consistent image quality under outdoor sunlight.

## **EPIC-2 — AI-Powered Disease Detection & Localization**

- **US-04:** As a farmer, I want the AI engine to detect visual signs of disease (fungal spots, blights, rusts, pest damage) in real time, so that I don't have to wait days for a lab diagnosis.
- **US-05:** As a field agronomist, I want the system to highlight affected leaf areas using bounding overlays or heatmaps, so that I can visually verify the exact infected areas.
- **US-06:** As a data scientist, I want the ability to retrain and deploy updated disease classification models with user-validated field samples, continuously improving accuracy.

## **EPIC-3 — Severity Scoring & Treatment Recommendation Engine**

- **US-07:** As a farm manager, I want detected diseases to be automatically classified with symptom details, so that I can apply targeted interventions immediately.
- **US-08:** As a farmer, I want the system to calculate disease severity percentage and issue specific pesticide/fertilizer dosage recommendations, avoiding chemical overuse.
- **US-09:** As an agronomy expert, I want to manually review and override AI diagnoses in edge cases, maintaining user trust and training data integrity.

## **EPIC-4 — Real-Time Monitoring, Outbreak Mapping & Alerts**

- **US-10:** As a farm supervisor, I want an interactive map dashboard showing disease hotspots across different field sectors, so that I can direct field workers efficiently.
- **US-11:** As a farmer, I want push notifications when micro-climate weather conditions favor disease spreading in my area, allowing me to apply preventive sprays.
- **US-12:** As a plant manager, I want to view historical crop disease progression across seasons and crop rotations to refine future planting strategies.

## **EPIC-5 — Analytics, Yield Forecasting & Historical Reporting**

- **US-13:** As an agricultural officer, I want automated weekly crop health summary reports, helping me evaluate farm performance without manually building spreadsheets.
- **US-14:** As a farm administrator, I want to export disease diagnostic reports to PDF/Excel format to present to insurers and government auditors.
- **US-15:** As a data analyst, I want predictive analytics that blend weather forecasts, soil sensor readings, and crop health metrics to forecast prospective crop yields.

## **EPIC-6 — System Administration, Offline Mode & Integration**

- **US-16:** As a system administrator, I want role-based access control (RBAC), ensuring seasonal workers, agronomists, and managers access only authorized field data.
- **US-17:** As an IT integration engineer, I want REST/GraphQL APIs connecting system alerts to smart IoT irrigation systems and farm equipment for automated response.

- **US-18:** As a system administrator, I want farm data to be securely encrypted during transmission and storage, protecting agricultural intellectual property.

### 3. Acceptance Criteria (Sprint 1 Selections)

#### US-01 — Automatic Leaf Photo Capture

- **Scenario 1: Automated shutter trigger on target alignment**

Given the user has opened the camera scanner and pointed the mobile device at a crop leaf,  
When the internal frame processor detects proper distance, clarity, and lighting,  
Then the app automatically captures the image without requiring manual shutter tapping.

- **Scenario 2: Blurred leaf rejection**

Given the mobile app camera is active under shaky hand movement,  
When a frame is analyzed as blurred (< 70% sharpness index),  
Then the system prompts the user with 'Hold Steady — Adjusting Focus' and suppresses photo capture.

#### US-02 — Edge Image Preprocessing

- **Scenario 1: Glare reduction and normalization**

Given a raw leaf photo captured under harsh sunlight,  
When the image enters the edge processing pipeline,  
Then the application normalizes color balance, reduces glare spots, and resizes the image to 512x512 px before cloud transmission.

- **Scenario 2: Low network connectivity optimization**

Given mobile network signal strength is low (2G/EDGE),  
When an image is prepared for cloud submission,  
Then the edge module applies lossy compression to minimize payload size while keeping leaf symptom features visible.

#### US-04 — Real-Time AI Crop Disease Detection

- **Scenario 1: Accurate identification of disease symptoms**

Given a preprocessed image containing active Early Blight or Leaf Rust symptoms is submitted,  
When the CNN model finishes inference,  
Then the system tags the crop condition as diseased, identifies the disease category with  $\geq 90\%$  confidence, and displays the result within 1 second.

- **Scenario 2: Healthy crop verified**

Given an image of a completely healthy leaf is submitted,  
When the model completes inference,  
Then the system identifies the status as 'Healthy' with zero false positive disease flags.

#### US-05 — Disease Location Highlighting

- **Scenario 1: Visual bounding box display**

Given the AI model has flagged an image as diseased,  
When the user views the diagnostic results screen,  
Then the app renders bounding boxes/heatmaps surrounding all identified lesion regions on the leaf photo.

## 4. Product Backlog Prioritization

### 4.1 Full Product Backlog Table

| ID    | Epic   | User Story Summary                    | Priority    | Value | SP | Dependency |
|-------|--------|---------------------------------------|-------------|-------|----|------------|
| US-01 | EPIC-1 | Automatic Leaf Photo Capture          | Must Have   | 8     | 5  | None       |
| US-02 | EPIC-1 | Edge Image Preprocessing              | Must Have   | 7     | 5  | US-01      |
| US-03 | EPIC-1 | Camera & Environment Calibration      | Should Have | 6     | 3  | US-01      |
| US-04 | EPIC-2 | Real-Time AI Crop Disease Detection   | Must Have   | 10    | 13 | US-02      |
| US-05 | EPIC-2 | Disease Location Highlighting         | Must Have   | 8     | 5  | US-04      |
| US-06 | EPIC-2 | Retrain AI Model with Field Data      | Should Have | 6     | 8  | US-04      |
| US-07 | EPIC-3 | Automatic Disease Classification      | Must Have   | 9     | 8  | US-04      |
| US-08 | EPIC-3 | Severity Scoring & Treatment Plan     | Must Have   | 9     | 5  | US-07      |
| US-09 | EPIC-3 | Manual Override of Diagnosis          | Should Have | 6     | 3  | US-07      |
| US-10 | EPIC-4 | Live Field Health Map Dashboard       | Must Have   | 8     | 8  | US-07      |
| US-11 | EPIC-4 | Weather & Disease Alert Notifications | Must Have   | 9     | 5  | US-10      |
| US-12 | EPIC-4 | Historical Disease Outbreak View      | Should Have | 6     | 5  | US-10      |
| US-13 | EPIC-5 | Automated Weekly Crop Reports         | Should Have | 6     | 5  | US-10      |
| US-14 | EPIC-5 | Export Reports to PDF/Excel           | Could Have  | 5     | 3  | US-13      |

|       |        |                                   |             |   |    |       |
|-------|--------|-----------------------------------|-------------|---|----|-------|
| US-15 | EPIC-5 | Predictive Yield Analytics        | Could Have  | 7 | 13 | US-13 |
| US-16 | EPIC-6 | Role-Based Access Control (RBAC)  | Must Have   | 8 | 5  | None  |
| US-17 | EPIC-6 | Smart Hardware / FMIS Integration | Should Have | 7 | 8  | US-13 |
| US-18 | EPIC-6 | Data Encryption in Transit & Rest | Must Have   | 9 | 5  | US-16 |

## 5. Sprint Backlog Planning

### 5.1 Sprint 1 Backlog Summary

| ID    | User Story Summary                  | Story Points | Priority         |
|-------|-------------------------------------|--------------|------------------|
| US-01 | Automatic Leaf Photo Capture        | 5            | Must Have        |
| US-02 | Edge Image Preprocessing            | 5            | Must Have        |
| US-03 | Camera & Environment Calibration    | 3            | Should Have      |
| US-04 | Real-Time AI Crop Disease Detection | 13           | Must Have        |
| US-05 | Disease Location Highlighting       | 5            | Must Have        |
| TOTAL | Sprint 1 Total Commitment           | 31 SP        | Target: 30–32 SP |

### 5.2 Task Breakdown

| Story ID | User Story                   | Development Tasks                                                                                                                                                                                                                                                                                                |
|----------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-01    | Automatic Leaf Photo Capture | <ul style="list-style-type: none"> <li>Integrate mobile camera SDK with frame-capture loop.</li> <li>Implement auto-focus/distance detection logic.</li> <li>Build local offline caching mechanism.</li> <li>Unit test frame capture under variable sunlight.</li> </ul>                                         |
| US-02    | Edge Image Preprocessing     | <ul style="list-style-type: none"> <li>Implement noise reduction and color balance filter.</li> <li>Build automatic cropping and normalization (512x512 px).</li> <li>Write adaptive compression routine for low bandwidth.</li> <li>Execute integration test: Capture -&gt; Preprocess -&gt; Upload.</li> </ul> |

|       |                                |                                                                                                                                                                                                                                                                                                                   |
|-------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-03 | Camera Calibration             | <ul style="list-style-type: none"> <li>• Design camera parameter configuration schema.</li> <li>• Build admin configuration screen for lighting/exposure.</li> <li>• Add parameter validation and fallback error messaging.</li> <li>• Persist parameter updates locally across app restarts.</li> </ul>          |
| US-04 | Real-Time AI Disease Detection | <ul style="list-style-type: none"> <li>• Curate and augment leaf dataset (blights, rusts, healthy).</li> <li>• Train baseline CNN / YOLOv8 computer vision model.</li> <li>• Containerize and deploy as low-latency microservice.</li> <li>• Benchmark inference latency against &lt; 1 second budget.</li> </ul> |
| US-05 | Disease Location Highlighting  | <ul style="list-style-type: none"> <li>• Extract bounding box coordinates from ML model output.</li> <li>• Build overlay rendering component in the UI.</li> <li>• Support rendering multiple lesion spots per frame.</li> <li>• Conduct usability testing with chief agronomy lead.</li> </ul>                   |

## 6. Sprint Goal and Metadata

### 6.1 Sprint Goal

*Sprint 1 Goal: Establish the foundational image acquisition and AI disease-detection engine, enabling the mobile platform to automatically capture leaf photos in the field, execute edge preprocessing, run real-time inference using a baseline AI model, and visually display bounding overlays on detected leaf disease lesions for field validation.*

### 6.2 Sprint Metadata

| Attribute                | Value                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Sprint Duration          | 2 Weeks (10 working days)                                                                                                         |
| Sprint Number            | Sprint 1 of 6                                                                                                                     |
| Team Composition         | 1 Product Owner, 1 Scrum Master, 2 AI/ML Engineers, 2 Developers, 1 QA Tester                                                     |
| Assumed Team Velocity    | 30–32 Story Points per sprint                                                                                                     |
| Sprint Commitment        | 31 Story Points (5 User Stories across EPIC-1 and EPIC-2)                                                                         |
| Definition of Done (DoD) | Code peer-reviewed & merged; unit/integration tests passing; acceptance criteria verified; demoed in Sprint Review; docs updated. |